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SBENT-2N

Simple and Compound Interest

(QUIZ#2)

I. Instructions: Solve each item. Show your solution if required.

A. Simple Interest:

1. Find the simple interest on Php5,000 at 6% per annum for 3 years. $I = 5000 \times 0.06 \times 3$
 $= \text{Php } 900$
2. What is the total amount after 4 years if Php2,500 is invested at 5% simple interest?
 $I = 2500 \times 0.05 \times 4 = 500$
3. A loan of Php1,200 earns Php216 interest in 3 years. What is the rate?
a. $R = I/Pt = 1200 \times 3216 = 0.06$ or 6%
4. How long will it take for Php800 to earn Php160 at 4% simple interest? $T = Pr / I =$
 $800 \times 0.04160 = 5$ years
5. If the interest earned is Php450 at 7.5% for 2 years, what is the principal? $P = rt/I =$
 $0.075 \times 2450 = \text{Php}3,000$

6. A principal of Php1,500 is invested at 8% for 1.5 years. Find the interest. $I = 1500 \times 0.08 \times 1.5 = \text{Php } 180$

7. What is the total amount after 2.5 years if Php2,000 is invested at 6%?

a. $= 2000 \times 0.06 \times 2.5 = 300$ $A = 2000 + 300 = \text{Php } 2,300$

8. How much interest is earned on Php3,000 at 5% for 4 years? $I = 3000 \times 0.05 \times 4 = \text{Php } 600$

9. A student borrowed Php2,400 and paid back Php2,760 after 3 years. What was the interest rate? $I = 2760 - 2400 = 360$ $r = \frac{360}{2400 \times 3} = 0.05$ or 5%

10. Find the principal if the total amount after 5 years at 4% is Php2,600.

a. $A = P(1 + rt)$ $2600 = P(1 + 0.04 \times 5)$ $2600 = P(1.2)$

b. $P = \frac{2600}{1.2} \approx \text{Php } 2,166.67$

c. Compound Interest

11. Find the compound amount of Php1,000 at 5% compounded annually for 3 years. $A = 1000(1 + 0.05)^3 = 1000(1.157625) = \text{Php } 1,157.63$

12. What is the compound interest on Php2,000 at 6% for 2 years?

$A = 2000(1 + 0.06)^2 = 2000(1.1236) = 2247.20$ $I = 2247.20 - 2000 = \text{Php } 247.20$

13. A principal of Php1,500 is invested at 4% compounded annually for 4 years. Find the total amount. $A = 1500(1 + 0.04)^4 = 1500(1.16985856) \approx \text{Php } 1,754.79$

14. How much interest is earned on Php3,000 at 5% compounded annually for 2 years?

$A = 3000(1 + 0.05)^2 = 3000(1.1025) = 3307.50$ $I = 3307.50 - 3000 = \text{Php } 307.50$

15. Find the compound amount of Php2,500 at 3% for 5 years. $A = 2500 (1 + 0.03)^5 = 2500 (1.15927407) \approx \text{Php } 2,898.19$

16. What is the compound interest on Php1,200 at 6% for 3 years?
 $A = 1200 (1 + 0.06)^3 = 1200 (1.191016) = 1429.22$ $I = 1429.22 - 1200 = \text{Php } 229.22$

17. A principal of Php4,000 is invested at 7% compounded annually for 2 years. Find the total amount. $A = 4000 (1 + 0.07)^2 = 4000 (1.1449) = \text{Php } 4,579.60$

18. How much interest is earned on Php2,200 at 4% compounded annually for 3 years?
 $A = 2200 (1 + 0.04)^3 = 2200 (1.124864) = 2474.70$ $I = 2474.70 - 2200 = \text{Php } 274.70$

19. Find the compound amount of Php1,800 at 5% for 4 years. $A = 1800 (1 + 0.05)^4 = 1800 (1.21550625) \approx \text{Php } 2,187.91$

20. What is the compound interest on Php2,000 at 8% for 2 years?
 $A = 2000 (1 + 0.08)^2 = 2000 (1.1664) = 2332.80$ $I = 2332.80 - 2000 = \text{Php } 332.80$

II. Directions: Do what is asked.

1. Calculate the present amount of Php40,000 due in 4 years at the given rate of interest

a. 6% compounded semi-annually

$n = 2, r = 0.06, t = 4$ $P = (1 + 2 / 0.06)^{2 \times 4} / 40000 = (1.03)^8 / 40000 \approx 1.26677 / 40000 \approx \text{Php } 31,576.85$

b. 8% compounded quarterly

$n = 4, r = 0.08, t = 4$ $P = (1 + 4 / 0.08) 4 \times 4 / 40000 = (1.02) 16 / 40000 \approx 1.37279 / 40000 \approx$
Php 29,138.86

c. 7% compounded monthly

$n = 12, r = 0.07, t = 4$ $P = (1 + 12 / 0.07) 12 \times 4 / 40000 \approx 1.32205 / 40000 \approx$ Php 30,255.74

d. 9% compounded daily.

$N = 365, r = 0.09, t = 4$ $P = (1 + 365 / 0.09) 365 \times 4 / 40000 \approx 1.43329 / 40000 \approx$ Php
27,907.64

2. A debt of Php8,000 will mature in four years. By assuming that the money is worth 9% compounded quarterly, calculate

a. the present value of this debt

$n = 4, r = 0.09, t = 4$ $P = (1 + 4 / 0.09) 4 \times 4 / 8000 = (1.0225) 16 / 8000 \approx 1.42762 / 8000 \approx$ Php
5,603.63

b. the value of this debt at the end of the two years $t = 2$

$A = 5603.63 \times (1.0225) 4 \times 2 = 5603.63 \times (1.0225) 8 \approx 5603.63 \times 1.19483 \approx$ Php 6,695.34

c. the value of this debt at the end of five years. $T = 5$